

PRAKASH RANJAN

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Profile Summary

Results-driven Software Engineer with 15+ years of experience in designing scalable, high-performance solutions across diverse domains. Proficient in Java, Spring Boot, Hibernate, RESTful APIs, and modern technologies, with expertise in system optimization and application development.

Skills

Languages: Java, Python, Typescript, SQL, Shell script

Technologies, Tools & Frameworks: Spring, Angular, Apache MQ, Quartz, REST, Micro-services, Tomcat, SNMP, Wsman, Redfish API, vmware VCenter, virtualization.

Container Technologies: Docker, Kubernetes.

Experience - 15 Years

Dell Technologies

Oct 2013– Present

Principal Software Engineer

Bangalore

- Developed a common shared library DARIO which was used by multiple teams which resulted in reduced customer call.
- Orchestrated parallel development with VMware, enabling feature support from day one.
- Collaborated with cross-functional teams to identify and mitigate project risks, resulting in reduction in project-related issues by 50 percent.
- Java, REST, Python, SQL, Typescript, Apache Tomcat, Apache MQ, Micro-service, Wsman, Redfish API, virtualization.

Samsung

Dec 2011– Oct 2013

Software Engineer

Bangalore

- Worked on python-based project which used to automate execution and generation of report for different test suits.
- Worked on an automation tool which used to deploy on various OS on physical servers with zero touch.
- Python, Perl, C, Lenaro

Procsys

Mar 2010– Dec 2011

Software Engineer

Bangalore

- Worked on developing a Gigabit Passive Optical Network based project.
- Worked on creating different shared libraries between different modules based on message queue.
- C, Linux POSIX Threads, Message queue.

Education

Siddaganga Institute Of Technology, Tumkur

Jul 2005 – Jul 2009

Electronics and Communication engineering, CGPA: 7.04/10

Achievements

- Received Dell champions award for implementing a modular architecture which helps in reducing future development execution effort by 75%.
- Led a parallel development initiative with VMware, involving multiple rounds of design discussions and improvements, ensuring feature support from day one.
- Collaborated with cross-functional teams to identify and mitigate project risks, resulting in reduction in project-related issues and ensuring seamless project delivery.

Hackathons

Use BMC to identify the network configuration drift in a NSX attached virtual distribution switch | In an NSX managed distributed switch system or in micro-segmented architecture which enables users to create a firewall for a group of virtual machines from different hosts (select pool of VM's Horizontally) or within a host. The creation of micro-segment involves several components like, Distributed switch, virtual network, Port group, VLAN ID etc. **2023**

Automating the detecting probable wrong hard disk | develop a solution for detecting faulty servers using historical performance data. Designed and implemented a system to analyze patterns and identify potential server issues. Reduced server downtime by identifying faults with over 85% accuracy. Delivered a scalable and efficient solution to support proactive server maintenance strategies. **2021**

Projects Work

Open Manage Enterprise Vmware Plugin | Solution to facilitate seamless automatic discovery, template based configuration, OS deployment and firmware update of Dell Servers at large scale. It works as a Cent-OS based plugin integrated into Vmware vCenter. **2021**

- Developed Proactive High availability to reduce the downtime of Dell servers in clusters and datacenter.
- Developed Proactive hardware manager based on SNMP alerts to predict the storage critical errors.

Open Manage Integration for Vmware Vcenter: | Solution to facilitate seamless automatic discovery, template based configuration, OS deployment and firmware update of Dell Servers at large scale. It works as a Cent-OS based plugin integrated into Vmware vCenter **2017**

- Involved in developing firmware update framework to schedule server firmware update from different level.
- Involved in developing alerting mechanism based on SNMP protocol to support monitoring os the Dell servers.

Dell API Migration Facilitator | Software sub-components communicate via APIs, which may require keys for secure access. Changes in APIs or keys require consuming programs to update for continued compatibility. DARIO help different software's to acheive it **2017**

- Worked as a lead for this project to facilitate the fetching of secret API key without releasing the new version of the product.

Dell Plugin for Nagios Monitoring tool | Dell plugin for Nagios is used to monitor and manage the dell server based on SNMP alerting. It uses SNMP and wsman protocol to communicate the Dell servers and pull the data. **2014**

- Involved in developing alerting mechanism based on SNMP protocol to support monitoring os the Dell servers.

Dell Plugin for HP Operational Manager | Dell plugin for HP Operations Manager to enable seamless integration and monitoring. Automated hardware alert handling, reducing manual intervention by 30%. Enhanced system reliability by providing real-time insights and streamlined operations. **2013**

- Involved in developing a framework for applying dell network policies across dell servers in HP operational manager environment.

Hardware automation test suite | Developed a hardware automation test suite in Python, reducing manual effort by 50 % and cutting test execution time by 40%. Integrated hardware control libraries and automated 100+ test cases for comprehensive coverage. Enhanced efficiency with seamless execution, detailed logging, and automated result analysis. **2012**

- This was a python based project to create a automation suite to run the multiple test cases of android os and capture the result in excel and create a report from it.

Linaro Automated Validation Architecture | LAVA is a automation testing framework used for scheduling and automating test jobs and storing the results which could be displayed on any browser by exposing through any web service **2012**

- Worked on automating the Linux deployment on Lenaro board and run the test suits automatically with zero touch.

Gigabit Passive Optical Network | Gigabit Passive Optical Network (GPON) is a fiber-optic technology delivering high-speed internet, voice, and video services using a point-to-multipoint architecture. It ensures scalable, cost-effective broadband access with data rates up to 2.5 Gbps downstream and 1.25 Gbps upstream. **2010**

- Mostly worked on creating a shared library based on message queue and linux posix thread.